Working paper, November 2026
Draft: comments welcome

Expressive Motivations in Committees

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Why do committee members with little information potentially harm the quality of a committee's decision by speaking? We run laboratory experiments where subjects can acquire information about the correct state of the world, communicate, and vote (with the goal of choosing the correct alternative). Contrary to the simple game theory prediction that subjects should truthfully report their information to the group, subjects frequently report information after not gathering any. Using a logit quantal response equilibrium we find that observed behavior is consistent with a desire to report signals to the group (even if the report is untrue) and a desire to tell the truth. (JEL C92, D71, D83)

Small committees are frequently used to make decisions under uncertainty. Juries try to determine the innocence or guilt of a defendant, hiring committees try to select the best candidate, and the Fed tries to make good monetary policy decisions. Committees where their members want to make the correct decision are valuable because they pool information from their members, achieving a better solution than if one person had made the decision. A committee's value might be diminished because of the reputational incentives that their group members experience. For instance, members that know their report to the committee will be used to evaluate their competence say less than they know to avoid being clearly wrong (Ottaviani and Sørensen 2006; Levy 2007), copy what others say (Scharfstein and Stein 1990; Prendergast 1993), or give an uninformative report when it is their turn to speak (Ottaviani and Sørensen 2001; Visser and Swank 2007). Even if real reputational motivations do not exist, there may still be an expressive motivation to be heard by the committee. Then, even a committee designed to minimize issues from reputational incentives may still face herding and groupthink.

In this paper we investigate committee reporting behavior when members can gather individually costly information and freely report to the group. We build a model that adds, to the basic instrumental utility, a utility to report to the group and a utility to be sincere in their report. We investigate how well this model fits choice data generated in a laboratory committee experiment. We believe that lab experiments are the best way to empirically evaluate our model because they allow us to observe behavior in an environment where information is expensive relative to its value, members are anonymous, and it is clear when members have no information. Expensive information

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minimizes the real benefit of buying a signal (instrumental motivations), anonymity allows us to eliminate reputation, and because we can observe who is uninformed it is easy to identify those reporting to the group with no valuable information. Jointly, this allows us to observe behavior that largely stems from expressive motivations.

The foundation of our choice model is a Condorcet jury: committee members have common preferences and make a group decision between two alternatives. Past theoretical literature has shown that when gathering information is individually costly, gathering information is a public good whose provision is a strategic substitute (Mukhopadhaya 2003; Persico 2004; Martinelli 2006, 2007; Gerardi and Yariv 2008; Koriyama and Szentes 2009). Our choice model combines the public good provision with an individual's psychological utility from reporting and from being sincere.

We use our choice theory to understand our model's identification strategy. In particular, our model classifies withholding information and reporting the opposite of information as mistakes. It expects buying and reporting that information when the value of providing the public good is large or there is a large utility from reporting and being sincere in that report. Not buying a signal and staying quiet is expected when the value of the public good is low, the utility from reporting is low and the utility to be sincere is high. On the flip side, when the benefit from the public good is small, the utility from reporting is large, and the utility from being sincere is small, we expect subjects to report having information despite not buying any. Our model of choice will identify reporting utility and sincerity utility that most closely match observed behavior.

To generate choice data we run an experiment where subjects choose to buy individually costly information, communicate, and ultimately vote for one of two alternatives. We observe subjects that did not acquire information report a signal between 30% and 49% of the time. We also find that subjects pollute the group's information by uninformed reporting less often when the report tally is weak, which is when an additional signal is worth the most. This is suggestive evidence in favor of our expressive model of behavior.

To evaluate whether this behavior stems from a utility from reporting and from being sincere, we compare the fit of a basic quantal response equilibria against the fit of a homogeneous quantal response with expressive and sincerity motivations on our data. We find a much better fit from our proposed model than the basic quantal response model (even when adjusting for the extra parameters we add). We show that this result is robust to several specifications.

Section I reviews the literature. Section II builds our identification strategy by laying out the predictions of our choice theory. Section III presents our experimental design. We present aggregate behavior data about our experiment in Section IV. We compare the fit of our expressive LQRE model with the simple LQRE model in Section V. Section VI concludes.

I. Literature Review

Expressive Motivations. Our central behavioral assumption comes from the costly voting literature. Because a single vote is almost never decisive in a large election, instrumental models struggle to explain turnout. A long tradition explains voting

despite low instrumental benefits by a direct benefit from the act itself: a sense of civic duty (Riker and Ordeshook 1968), the expression of a preference (Brennan and Lomasky 1993; Hamlin and Jennings 2011), or social pressure (Gerber, Green, and Larimer 2008; DellaVigna et al. 2017). Recent work adds expressive payoffs from voting to committee models with exogenous information and shows that they can undermine information aggregation and truthful communication (Morgan and Várdy 2012; Midjord, Rodríguez Barraquer, and Valasek 2021; Breitmoser and Valasek 2024). We find behavior consistent with just expressive motivations in our No Communication condition. We extend this paradigm by attaching the expressive utility to reporting and adding a motivation to report sincerely. This helps explain behavior in our Signal Sharing and Sequential Buying conditions, where subjects choose to report before voting.

Committees with Costly Information. When information is costly, a committee member's information is a public good. Because every additional informed member makes each member less likely to be the decisive vote, information acquisition is a strategic substitute (Mukhopadhaya 2003; Persico 2004; Martinelli 2006, 2007; Gerardi and Yariv 2008; Koriyama and Szentes 2009). Großer and Seebauer (2016), Bhattacharya, Duffy, and Kim (2017), and Elbittar et al. (2020), in committees that cannot communicate, test a sharp prediction of this theory: members of smaller committees should buy signals more often, because they are more likely to be decisive. They find that subjects in smaller committees do not acquire information more often.[1] We find an insensitivity along a similar margin. The instrumental predictions differ across our conditions (4.6% in No Communication and Signal Sharing, 20% in Sequential Buying), yet subjects buy signals at nearly identical rates (between 37% and 41%) in all three conditions, far above those predictions. Our model offers an explanation: subjects buy a signal because they want to report it, so the purchase decision is mostly disconnected from how beneficial a member's signal is.

Herding and Reputation. People in groups who observe others' choices may rationally set aside their own information, either because earlier choices outweigh it (Banerjee 1992; Bikhchandani, Hirshleifer, and Welch 1992) or because following others protects their reputation (Scharfstein and Stein 1990). A theoretical literature shows how reputational concerns lead committee members and experts to distort what they say and how they vote (Ottaviani and Sørensen 2001; Levy 2007; Visser and Swank 2007), and in experiments where members are judged by their reports, subjects rarely misreport (Fehrler and Hughes 2018; Renes and Visser 2025). Our design removes these reputational incentives and still observes frequent distortion: some members fabricate signals, while others acquire and share information, benefiting the group. A desire to report and a desire to be sincere explain both.

Lying in Committees. We also contribute to the experimental literature on lying in committees. When signals are handed to subjects in common-value committees, they tend to report them truthfully (Guarnaschelli, McKelvey, and Palfrey 2000; Goeree and Yariv 2011). When subjects choose whether to acquire information, the

[1] Bhattacharya, Duffy, and Kim (2017) do find that bigger groups buy signals at a lower rate when signals are perfectly informative. This is the only combination of parameters in the three papers that rejects the null of equal purchase rates across committee sizes.

evidence diverges: Mukherjee et al. (2024) find that voters in a laboratory social media environment frequently misreport the strength of their signal when signal strength is a choice, while Bandyopadhyay et al. (2024) find that subjects who make a binary decision to buy a signal rarely lie. In our binary signal buying decision, we find frequent lying: between 30% and 49% of members who did not buy a signal report one anyway.

II. Theory

There are n committee members who want to choose p when the state of the world, ω, is Purple and choose g when the state of the world is Green. Each member earns 1 if the group choice matches the state and 0 otherwise (we normalize the payoff of being correct to 1).[2] Players all begin at a symmetric prior of the two states of the world, $P_0(\omega = \text{Purple}) = P_0(\omega = \text{Green}) = 0.5$. Players can pay a cost, $c > 0$, to acquire a signal, ς_i, with strength, $x > 0.5$ (ρ indicates that Purple is more likely while γ indicates that Green is more likely).[3] Signals are independent conditional on the state of the world, ω. Call $k = |\#\rho - \#\gamma|$ the true signal tally. Then, $P(\text{majority signals correct}) = x^k/(x^k + (1-x)^k)$. Communication takes place between the acquisition of signals and voting. To communicate a player will send a message to the group, m_i (this is a cheap talk message). After all signals are bought and communication is done, each group member chooses to {vote p, vote g, or abstain}. The group choice is the color that gets the majority of the votes (if there is a tie, a random color is chosen).

Messages will fall into different classes. We will consider messages that communicate a signal $m \in \{\rho, \gamma\}$ and class other messages as $m = \emptyset$. Call h_i the history of signal reports. We will call a signal report *sincere* if the report matches the player's signal ($m_i = \varsigma_i$). If players report sincerely, we can sum the messages from h_i and we will get the true k. But if there is insincere reporting, players will evaluate the signals based upon the probability that each player in the history truthfully reported.

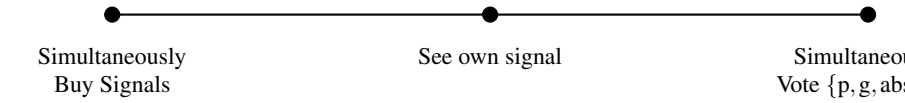


FIGURE 1. SEQUENCE OF DECISIONS IN THE NO COMMUNICATION CONDITION

Players in the No Communication condition buy information without seeing others' behavior.

[2] This is without loss of generality: imagine that the group wins a prize of Prize for being correct and has a cost of Cost for buying a signal, then the players face the same problem if we transform the payoff into 1 and the cost into Cost/Prize.

[3] More formally, when the state is Green, the committee member will get the signal γ with probability x and ρ with probability $1-x$. When the state is Purple, the committee member will get the signal ρ with probability x and γ with probability $1-x$. Signals are independent conditional on the state.

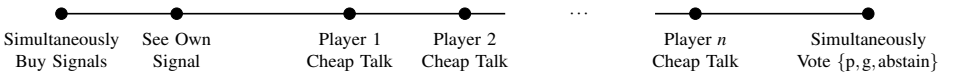


FIGURE 2. SEQUENCE OF DECISIONS IN THE SIGNAL SHARING CONDITION

Players in the Signal Sharing condition also buy signals simultaneously, without seeing others' behavior, but then share their information sequentially before voting.

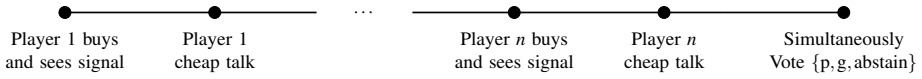


FIGURE 3. SEQUENCE OF DECISIONS IN THE SEQUENTIAL BUYING CONDITION

Players in the Sequential Buying condition buy and share signals one at a time, so players can influence one another's buying behavior.

A. Strategic Substitutes

We argue that buying signals is a strategic substitute in the different conditions. For now, we will talk about behavior under purely instrumental motivations (players just maximize the expected terminal accuracy of the group's judgment).[4] Once we have shown that buying signals is a strategic substitute we will add in our expressive reporting motivations.

Under purely instrumental motivations, players vote and report sincerely. Because our signals are equally precise and independent and the prior is symmetric, the posterior favors the color that has the majority of signals, so majority rule is the most efficient way to aggregate them and sincere voting is a Nash equilibrium (Austen-Smith and Banks 1996). Conditional on being pivotal, the others' ballots are tied, so a voter's own signal decides the outcome and they should vote for the color it indicates. A voter who did not buy a signal strictly prefers to abstain because their ballot carries no information. The vote can only turn an informed majority decision into a tie or a tie into an uninformed decision (Feddersen and Pesendorfer 1996). Sincere reporting follows the same logic. If everyone votes for the alternative that the reported signals support, an untruthful report leads to a worse decision whenever it changes which color the signals support, and is irrelevant otherwise, so reporting sincerely is weakly dominant. The result is that signals are translated into the group choice in the same way across conditions.

No Communication players that only consider instrumental motivations will only buy signals with the hope of making the group's decision better. Players will only improve the group's accuracy by being a decisive voter *and*, conditional on the information that others have, making the decision better. When other voters vote sincerely this happens only when the player's vote breaks a tied vote. The probability that a voter breaks ties is decreasing as other players buy signals and vote them. This is true for most symmetric mixed strategy Nash equilibria, breaking down only

[4] A more thorough analysis can be found in the appendix.

when the probability that each player buys a signal is very high (for our experimental parameters, above 84.1%). Our experiment has signal buying rates far lower than this.

Signal Sharing players will report their signals sincerely and vote for the alternative that is supported by the reported signals, flipping a coin to break ties. This replicates the aggregation of signals from the No Communication condition. This implies that information acquisition will be treated the same as in the No Communication condition: buying signals is a public good whose provision is a strategic substitute.

Sequential Buying players report in sequence. So, their incentive to buy a signal depends upon the observed reports of past players and the expected reports of future players. Players will report truthfully when buying and stay quiet when they do not buy. So, players will make decisions given k, the absolute difference in the color of the group's signals. In general, early players will put off buying signals because they can push the burden of buying signals onto later players. For different costs, there are different equilibria. The first equilibrium is for just the last player to buy a signal (this happens when the cost of a signal is high). The second equilibrium is for the third and second to last players to buy signals then, if their signals disagree, the last player buys a signal (this is for lower costs). For our experiment's parameters, $x = 0.8$ and $c = 0.25$, if players only care about instrumental benefits, only the last player will buy a signal. This gives the group an accuracy of 0.8. If the third and second to last players each buy a signal and the last player buys a signal if theirs disagree, then the group gets an accuracy of 0.896.[5] For this equilibrium our cost would need to be much lower ≈ 9.6 cents. To have more players buy signals there needs to be lower and lower costs (or as we will see, expressive motivations).

Instrumental theory predicts in the No Communication and Signal Sharing conditions that players will buy signals at a rate of 4.6% and in the Sequential Buying condition that the last player will buy a signal (buy rate of 20%). This translates to a group accuracy of 0.563 in No Communication and Signal Sharing and 0.8 in Sequential Buying. For more details on instrumental theory, see the appendix.

B. Expressive Behavior

Now, we add the expressive components to the utility function: r, a desire to report to the group, and s, a desire to report sincerely. The utility of players from choice C consists of the group's expected accuracy given that choice, $E[Acc|C]$, the psychological rewards enjoyed, r and s, and costs that are paid, c.

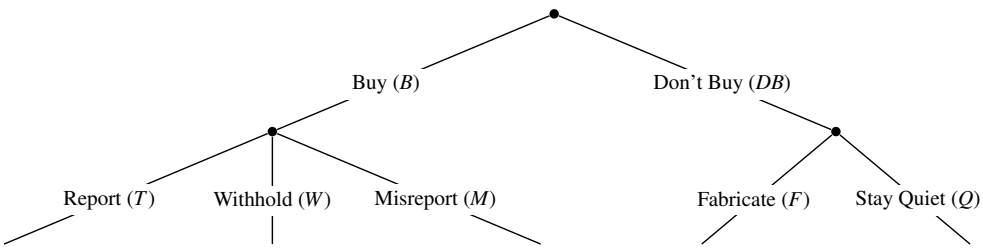


FIGURE 4. DECISION TREE

[5] The need to have a player "backstop" others in the event their signals cancel out is a consequence of the non-concavity in the value of these signals (Radner and Stiglitz 1984).

To structure the choice environment it is helpful to think about a decision as a nested choice: first the player chooses if they will buy a signal (B or DB), then they choose what to report. When a player buys a signal, they have the opportunity to report truthfully (T), report the opposite signal of what they observed (M), or withhold the signal (W). If the player does not buy a signal, they get to choose whether to report a signal (F) or report nothing (Q).

Players only see their signal after buying. Remember that No Communication players have no information about other players' choices at the time of buying and at the time of reporting. Signal Sharing players have only their player number when they buy signals and learn previous players' reports when they arrive to their reporting decision. Sequential Buying players know past players' reports when they buy and when they report.

Abusing notation, we hide the strategies of other players, σ_{-i}, the history of reports, h_i, and the signal, ς_i, calling $E[Acc|C]$ the expected accuracy for a player taking a reporting action. Then, the utility for a player is

$$U = \begin{cases} r+s+E[Acc|T]-c & \text{if Report Truthfully} \\ E[Acc|W]-c & \text{if Withhold} \\ r+E[Acc|M]-c & \text{if Misreport} \\ r+E[Acc|F] & \text{if Fabricate} \\ s+E[Acc|Q] & \text{if Stay Quiet} \end{cases} \tag{1}$$

We will define $E[\Delta(C_1,C_2)] = E[Acc|C_1]-E[Acc|C_2]$, the difference in expected accuracy of the group's decision between two decisions. In general, reporting truthfully will have at least as high an expected accuracy as other options. This is the only way that more information is added to the group. Withholding does as much as staying quiet does. Misreporting is harmful, taking the group further from the true state of the world. Fabricating a signal tends to cause less harm because players will choose to fabricate a signal that agrees with past signals. Staying quiet just carries the status quo forwards, with little effect on the group's decision.

First, we show that withholding and misreporting are dominated by the other alternatives. Reporting truthfully is better than withholding when

$$r+s > E[\Delta(W,T)] \tag{2}$$

This is generally true because the left hand side is positive, and the right hand side, because withholding adds no information to the group, is at most 0.

Reporting truthfully is better than misreporting when

$$s > E[\Delta(M,T)] \tag{3}$$

Misreporting is usually more harmful than withholding: while withholding just eliminates the benefit of adding a true signal to the tally, misreporting reduces the accuracy of the group's judgment. So, reporting truthfully is generally better than misreporting.

So, we can reduce a player's decision to whether they report truthfully (T), fabricate (F), or stay quiet (Q).

A player prefers reporting truthfully over fabricating when:

$$s > c - E[\Delta(T,F)] \tag{4}$$

the utility from being sincere is greater than the difference between the cost and the personal benefit of buying a signal over fabricating one.

A player prefers reporting truthfully over staying quiet when:

$$r > c - E[\Delta(T,Q)] \tag{5}$$

the utility from reporting is greater than the difference between the cost and the personal benefit of buying a signal over staying quiet.

A player prefers staying quiet over fabricating when

$$s - r > E[\Delta(F,Q)] \tag{6}$$

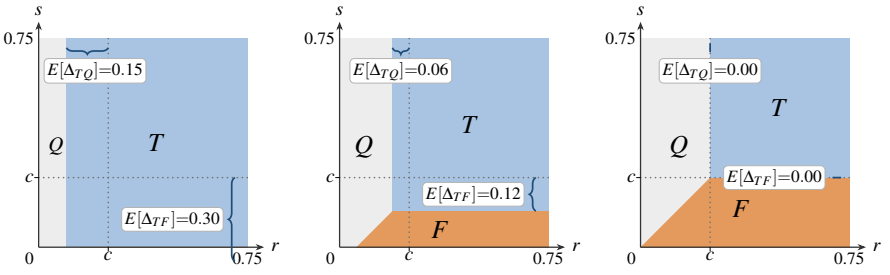
that is, the net utility from sincerity, $s-r$, is greater than the accuracy gained by fabricating over staying quiet. Equivalently, fabricating is preferred when $r - s > E[\Delta(Q,F)]$.

1. *No Communication and Sequential Buying*

No Communication does not have a report phase, it has a voting phase. We will apply the same paradigm to the voting phase: for example, if a player does not buy a signal and votes, we say that they fabricated.

We will present regions of the space of r and s where players will make different buy and report decisions. Players in the No Communication and Sequential Buying conditions only gain the information of a signal between buying and reporting their signals. Because this information will never influence players to not report truthfully, we can treat the choice at the buy level as whether to fabricate, stay quiet, or report truthfully. Figure 5 fixes $E[Acc|C]$ for some scenarios. It shows that high levels of s and r will motivate players to report truthfully and high levels of $E[Acc|T]$ will also lead to truthful reporting. This is the only decision that is made with real information, so the choice will tend to boost the group's accuracy by at least as much as fabricating and staying quiet.

When players in the Sequential Buying condition are at a history where the benefit from a truthful report is large, players will tend to report truthfully and sometimes stay quiet. The harm from fabricating will tend to be high here because these are the times when the extra signal has a large effect on the group's decision. Once this information has been committed, the group might not come back.

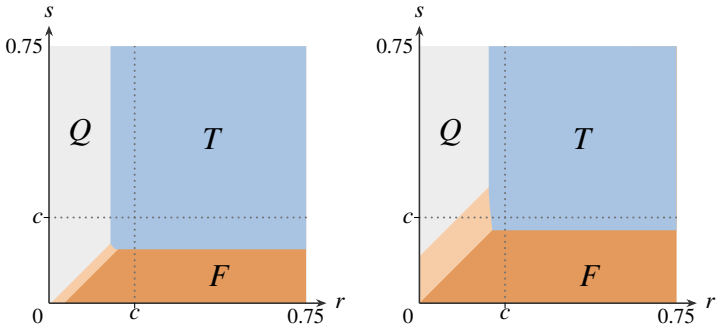


(A) LARGE IMPACT (B) SMALL IMPACT (C) NO IMPACT

FIGURE 5. DECISIONS BY TASTE PARAMETERS (NO COMMUNICATION AND SEQUENTIAL BUYING).

2. *Signal Sharing*

Players in the Signal Sharing condition will buy against an expectation of their opportunities in the buy and don't buy phases because they do not know the history of reports when they buy a signal. In general there is very little variation in the expected value of buying a signal for different seats. They solve the problem backwards: first they ask how players react to the possible range of $E[\Delta(F,Q)]$, then they ask how much better buying is as a result.



(A) PLAYER 3 (B) PLAYER 5

F after some histories, Q after some histories

FIGURE 6. DECISIONS BY TASTE PARAMETERS (SIGNAL SHARING).

The choice problem illustrated in Figure 6 is different for early and later players. Early players have a predictable history of reports, and therefore can predict their fabricate and stay quiet decision. In contrast, later players have a larger range of realized $E[\Delta(F,Q)]$, and consequently a larger range of possible behavior. There is also a slight decline in the expected benefit from buying a signal for players later in the order.

III. Experimental Design

We designed our experiment to closely mirror our theory. The one variable that we manipulate is the communication structure. Individual subjects only make their decisions under one communication condition.

We chose to have 5 player groups with a reward of \$1 when the group's decision matches the state of the world and a cost of 25 cents to buy a signal with a strength of 80%.[6] These numbers were chosen to provide a small incentive to buy a signal, allowing us to bias our sample towards decisions that are expressive.

[6] The instructions told subjects that each group member would be paid 1000 points if their group chose the correct color and that each signal cost 250 points. This is with a conversion rate of 1000 points to \$1.

TABLE 1—SESSION SUMMARY BY CONDITION

Condition	Number of Sessions	Number of Players
No Communication	4	55
Signal Sharing	4	50
Sequential Buying	4	50

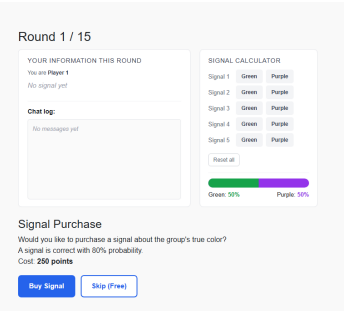
Each session had at least 10 inexperienced subjects (no subject took part in more than one session) playing the game for 15 rounds. At the beginning of each round, subjects were randomly formed into new groups.[7]

Table 1 shows the sessions run in each condition. We are still running sessions, with an ultimate goal of 105 subjects in each condition.

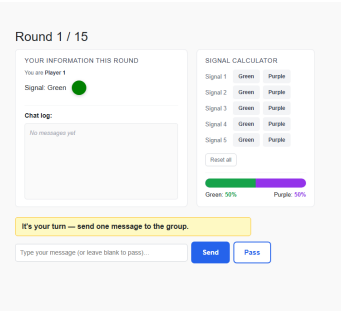
The sequence of our experiment is the same as in the theory. Each subject was given a $5 show-up fee. When a subject decides to buy a signal by clicking on the buy signal button, they are shown a ball drawn from a virtual bag. The virtual bags always have 8 balls of the correct color and 2 of the incorrect color (we also express this to them as a colored signal giving them an 80% chance of the correct color being the signal's color). If they are in a communication condition, they then get an opportunity to communicate with others. Figure 7 shows the beginning of a Sequential Buying round.

Communication takes the form of a message being sent from one player to all others; each player may send one free-text message to the group or pass. Subjects in the Signal Sharing condition take turns, going from player 1 to player 5 in order, until all subjects have either sent a message or passed. In the Sequential Buying condition, player 1 buys a signal, sees it and then sends a message to the group, then player 2 buys a signal, sees it, and sends a message to the group, this continues until all players have made their signal buying decision and messaged. In all conditions, once signal buying and messaging are done, all players simultaneously vote on the choice that they want their group to make. Players will not see each other's signals or even their choice to buy a signal unless that player reports it to the group. Apart from signal costs, every member's payoff depends only on whether the group chooses the correct color.

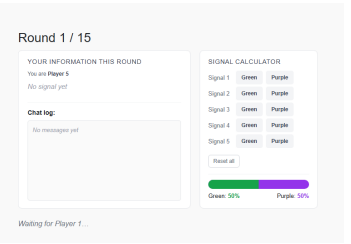
[7] This is common in experiments that want to test the predictions of a stage game but also give their subjects time to learn how to play the game.



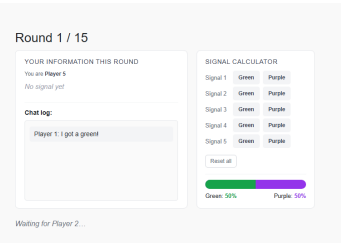
(A) PLAYER 1 DECIDES WHETHER OR NOT TO BUY A SIGNAL.



(B) PLAYER 1 BOUGHT A SIGNAL, AND NOW HAS TO SEND A MESSAGE TO THE GROUP.



(C) PLAYER 5 IS WAITING ON PLAYER 1 TO MAKE THEIR BUY DECISION AND TO SEND A MESSAGE.



(D) PLAYER 5 SEES PLAYER 1'S MESSAGE AND IS NOW WAITING ON PLAYER 2.

FIGURE 7. THE BEGINNING OF A SEQUENTIAL BUYING ROUND

Subjects can abstain from voting. The group's decision is made by majority rule: the color with the most votes wins and in the event of a tie a color is randomly chosen by the computer.

Subjects were given instructions before the game started (Sequential Buying instructions are included in Appendix A). We included a signal calculator for subjects to use which allows them to compute the Bayesian posterior given any combination of signals. The calculator can show them that the accuracy of the group after two confirmatory green signals is 94.12%. The intent was to show subjects how quickly returns to signals diminish without a heavy-handed explanation. The calculator can also show subjects that one green and one purple signal cancel out. The hope is to decrease any uncertainty that a subject might have about the actual math, making the whole task easier while maintaining the same structure that the game theory model has.

We had a short quiz at the end of the instructions to ensure that subjects understood the signal strength, the round structure, and the voting mechanism.

Subjects were undergraduate, masters, and PhD students at George Mason University. Experiments were conducted at the Interdisciplinary Center for Economic Science laboratory at George Mason's Fairfax campus. No subject was allowed to participate in more than one session of this experiment. So that subjects always made some money, subjects were paid the sum of their earnings from each round. Total earnings, including the $5 show-up payment, averaged about $15 ($14.72) for a 45

minute experiment. We obtained approval from the George Mason University IRB and ran our experiments on oTree (Chen, Schonger, and Wickens 2016).

IV. Data

We will present data about our experimental conditions. We present the share of player-rounds in which a signal is bought in our conditions, the mean accuracy of the groups' choices, the prevalence of fabricating signals after not buying, and the prevalence of misreporting and withholding of signals after buying. Throughout this section we will compare observed behavior to that predicted by a purely instrumental model.

A. Signal Acquisition

We look at the share of player-rounds in which a signal is bought. The instrumental game theory point prediction is that players buy signals 4.6% of the time in Signal Sharing and No Communication and 20% of the time in Sequential Buying.

TABLE 2—SIGNAL BUYING BY CONDITION

Condition	Signal Buy Rate	N (Participants)
No Communication	37.2% (4.3%)	55
Signal Sharing	41.3% (3.8%)	50
Sequential Buying	39.5% (4.0%)	50

Notes: Average share of rounds where a player bought a signal. Standard errors are in parentheses (clustered by participant).

Table 2 shows similar signal buying rates, between 37% and 41% in every condition. Under a cluster robust sandwich two-sided t-test, the three conditions do not have statistically significantly different rates of signal buying.[8] Observed purchase rates are far above the instrumental equilibrium predictions in every condition.

B. Accuracy

Subjects in each condition buy signals at similar rates. Even so, if subjects are taking advantage of their ability to coordinate on buying signals in the Sequential Buying condition, we might expect the Sequential Buying groups to be correct more often.

TABLE 3—GROUP ACCURACY BY CONDITION

Condition	Group Accuracy	N (Groups)
No Communication	72.7% (3.5%)	165
Signal Sharing	70.7% (3.7%)	150
Sequential Buying	80.7% (3.2%)	150

Notes: Share of rounds the group's decision matched the true state. Standard errors are in parentheses (clustered by group).

[8] No Communication vs. Signal Sharing: $p = 0.474$; Sequential Buying vs. No Communication: $p = 0.700$; Sequential Buying vs. Signal Sharing: $p = 0.734$.

Table 3 shows that Sequential Buying groups have a higher mean accuracy than those in the other conditions. Under a cluster robust sandwich two-sided t-test, the difference between Sequential Buying and Signal Sharing is significant at the 5% level and the difference between Sequential Buying and No Communication is significant at the 10% level but not at the 5% level. This is suggestive evidence that Sequential Buying groups are more accurate.[9]

This evidence is also consistent with our expressive theory: if players can see the impact of their fabrication and signal buying, this leads to higher accuracy.

C. Uninformed Voting

We will present what proportion of voters vote when informed and what proportion vote when uninformed. This is the natural analogue of our measure of insincere reporting in the next section.

TABLE 4—VOTING RATES

	Uninformed Voting	**Informed Voting**
No Communication	91.1% (472/518)	98.7% (303/307)

Notes: Uninformed voting is out of all the times that a signal was not bought; informed voting is out of all the times that a signal was bought.

Table 4 shows that voters vote while uninformed very often. This is much higher than in most treatments in Großer and Seebauer (2016), Bhattacharya, Duffy, and Kim (2017), and Elbittar et al. (2020). Our signals are much costlier than theirs. The closest comparison is the treatment in Bhattacharya, Duffy, and Kim (2017) with a very costly signal, where the uninformed voting rate is 72.8%.

D. Insincere Reports

We will first present how often players fabricate, misreport, and stay quiet. Then we will break down which situations these actions are more common in.

1. Text analysis

We use text analysis to classify subjects' chat messages into 3 categories: "I got a purple signal", "I got a green signal", and all other messages. Most messages are signal reports (85% of messages in Signal Sharing and 83% in Sequential Buying), and about 60% of all messages are a single word such as "purple", "green", "p", or "g". Many subjects stay quiet when they have bought no signals; some choose to message about the voting or tell others to buy signals.

Messages that say "vote green" are slightly unclear. This could be interpreted as a signal report, but we choose to classify these as a non-report.

[9] No Communication vs. Signal Sharing: $p = 0.686$; Sequential Buying vs. No Communication: $p = 0.095$; Sequential Buying vs. Signal Sharing: $p = 0.043$.

2. *Prevalence of Types of Reporting*

Once we have coded messages, we can classify any mismatch between the signal report and the actual signal as an insincere report.

TABLE 5—INSINCERE REPORTING RATES BY CONDITION

	Fabricating	**Withholding**	**Misreporting**
Signal Sharing	30.0% (132/440)	5.5% (17/310)	1.6% (5/310)
Sequential Buying	48.5% (220/454)	12.5% (37/296)	3.0% (9/296)

Notes: Fabricating rate is out of all the times that a signal was not bought; misreporting and withholding rates are out of player-rounds that bought a signal.

Table 5 shows that fabricating is the most common form of insincerity. Misreporting is rare (1.6% and 3.0%) and withholding is much less common than fabricating (5.5% and 12.5%). This is consistent with withholding and misreporting being mistakes.

3. *Timing of Fabrications*

Tables 6 and 7 report fabrication among non-buyers by a naive signal tally, $\sum h$, for Signal Sharing and Sequential Buying.

TABLE 6—FABRICATION BY SEAT AND k (SIGNAL SHARING)

Report Tally	P1	P2	P3	P4	P5
$\sum h = 0$ none	35.0% (80)	15.2% (33)	20.8% (24)	26.3% (19)	21.1% (19)
$\sum h = 1$	·	24.6% (57)	24.4% (41)	23.8% (42)	41.7% (36)
$\sum h = 2$	·	·	47.1% (17)	40.9% (22)	23.5% (17)
$\sum h = 3$	·	·	·	45.5% (11)	43.8% (16)
$\sum h = 4$	·	·	·	·	50.0% (6)

Notes: Percent of non-buyers who reported a color anyway; sample size in parentheses. $\sum h$ is the absolute net tally (green minus purple) of reports from lower-numbered teammates. A dot marks an impossible cell. Buying is simultaneous, so the distribution of $\sum h$ differs by seat.

TABLE 7—FABRICATION BY SEAT AND k (SEQUENTIAL BUYING)

Report Tally	P1	P2	P3	P4	P5
$\sum h = 0$ none	68.0% (50)	38.5% (13)	50.0% (8)	0.0% (3)[a]	28.6% (7)
$\sum h = 1$	·	43.4% (83)	43.6% (39)	35.7% (42)	25.7% (35)
$\sum h = 2$	·	·	57.5% (40)	48.3% (29)	53.1% (32)
$\sum h = 3$	·	·	·	78.6% (28)	18.2% (22)
$\sum h = 4$	·	·	·	·	78.3% (23)

Notes: As in Table 6. [a] fewer than 5 observations.

Fabrication tends to increase with the report tally. For the Signal Sharing condition, the number of non-buyers across cells follows a binomial-like distribution. This stems from buying in the Signal Sharing condition being blind to the tally that they will be at. In contrast, Sequential Buying subjects are not blind to the tally at the buy decision. We see that the cells at $\sum h = 0$ are sparse; Sequential Buying subjects choose to buy more often when $\sum h = 0$.

We more rigorously test whether a player that does not buy a signal is more likely to fabricate a signal when the tally of signals gets stronger. We estimate the logistic regression:

$$(7)\quad P\Big(\text{fabricate}_{mt}=1 \;\Big|\; \textstyle\sum h_{mt}, i_{mt}\Big) = \Lambda\Big(\alpha_0+\alpha_1\sum h_{mt}+\sum_{j=3}^{5}\ell_j\,\mathbb{1}\{i_{mt}=j\}\Big)$$

where $\Lambda(z)=e^z/(1+e^z)$, mt indexes participant m in round t, $\sum h_{mt}$ is the absolute net tally of reports from lower-numbered teammates, i_{mt} is the participant's player number (seat) in that round, the ℓ_j are seat fixed effects with seat 2 as the omitted category. We exclude $i=1$, because they only react to $\sum h=0$, not adding any useful information about the subjects' reaction to different $\sum h$. The sample is restricted to player-rounds in which the player did not buy a signal. We cluster standard errors by participant, since the same subject's tendency to fabricate is likely correlated across their rounds. One more confirming report is correlated with odds of fabricating that are 1.75 times larger in Sequential Buying ($\alpha_1=0.56$, SE $=0.19$, $p=0.003$) and 1.40 times larger in Signal Sharing ($\alpha_1=0.34$, SE $=0.14$, $p=0.019$).

4. *Withholding and Misreporting*

Tables 8 and 9 present withholding and misreporting as a function of the naive tally.

TABLE 8—MISREPORTING AND WITHHOLDING BY SEAT AND k (SIGNAL SHARING)

	P1			P2			P3			P4			P5		
Report Tally	M	W		M	W		M	W		M	W		M	W	
$\sum h=0$ none	1.4%	4.3%	(70)	0.0%	9.1%	(22)	0.0%	4.8%	(21)	6.2%	6.2%	(16)	0.0%	10.0%	(10)
$\sum h=1$	·			0.0%	5.3%	(38)	3.4%	3.4%	(29)	4.8%	0.0%	(21)	0.0%	0.0%	(20)
$\sum h=2$	·			·			0.0%	5.6%	(18)	0.0%	0.0%	(11)	0.0%	13.3%	(15)
$\sum h=3$	·			·			·			0.0%	25.0%	(8)	12.5%	0.0%	(8)
$\sum h=4$	·			·			·			·			0.0%	33.3%	(3)[a]

Notes: M: reported the opposite color; W: bought but sent no report. Signal buyers per cell in parentheses. A dot marks an impossible cell. [a] fewer than 5 observations.

TABLE 9—MISREPORTING AND WITHHOLDING BY SEAT AND k (SEQUENTIAL BUYING)

	P1			P2			P3			P4			P5		
Report Tally	M	W		M	W		M	W		M	W		M	W	
$\sum h=0$ none	1.0%	10.0%	(100)	0.0%	23.1%	(13)	3.8%	15.4%	(26)	9.1%	9.1%	(11)	0.0%	25.0%	(4)[a]
$\sum h=1$	·			2.4%	0.0%	(41)	4.3%	26.1%	(23)	6.7%	6.7%	(15)	16.7%	16.7%	(12)
$\sum h=2$	·			·			0.0%	7.1%	(14)	0.0%	11.8%	(17)	0.0%	37.5%	(8)
$\sum h=3$	·			·			·			0.0%	20.0%	(5)	20.0%	40.0%	(5)
$\sum h=4$	·			·			·				0.0%		0.0%		(2)[a]

Notes: As in Table 8. [a] fewer than 5 observations.

Misreporting is rare and seems uniformly distributed. This fits with the assertion that misreporting is a mistake. In Sequential Buying, withholding does seem to climb as the naive tally rises, casting some doubt on the assertion that it is a mistake.

E. Summary

We find evidence that buying rates in these three environments are almost identical. Players would report sincerely if they only cared about the instrumental motivations. We see frequent insincere reporting. In the Signal Sharing and Sequential Buying conditions we see that players fabricate frequently. They also tend to fabricate reports as the naive signal tally climbs. The ability to target buying behavior to the situations where $\sum h = 0$ is seen in the Sequential Buying condition, where players can change their buying behavior in response to the signal tally. This may explain the higher accuracy in the Sequential Buying condition; a greater ability to target distortions and contributions could improve accuracy further. In the next section, we handle these different information structures to better understand the choices that subjects made.

V. Structural Estimation

We will compare both a logit model with just instrumental motivations (a natural benchmark) and one with both expressive and instrumental motivations. We compare Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) to evaluate the fit of the models. These measures are commonly used to compare fit of models by taking the maximum likelihood of models and penalizing the models with too many parameters. In general, BIC is better at picking simple models that explain the underlying structure of reality (AIC is better at picking models that do well at forecasting). We find that our expressive motivations model has a much smaller AIC and BIC than our purely instrumental model (by more than 90 in every condition).

A. Estimation Overview

The variables of interest s and r affect both the lower level report decisions (players are more likely to withhold or misreport when s and r are low and more likely to fabricate when $r-s$ is high) and also the higher level buy decision (given your expectation of the lower level good decision, you should choose a different path).

We specify a logit choice model where players make their buy decision after seeing their preference shock ε_{Cm} at each of the lower level report decisions. This leads to a player level utility function:

$$U_m = \begin{cases} r+s+E[Acc|T]-c+\varepsilon_{Tm} & \text{if Report Truthfully} \\ E[Acc|W]-c+\varepsilon_{Wm} & \text{if Withhold} \\ r+E[Acc|M]-c+\varepsilon_{Mm} & \text{if Misreport} \\ r+E[Acc|F]+\varepsilon_{Fm} & \text{if Fabricate} \\ s+E[Acc|Q]+\varepsilon_{Qm} & \text{if Stay Quiet} \end{cases} \tag{8}$$

Players will look forward and create a plan given their drawn parameters, then execute that plan. In the No Communication and Signal Sharing conditions, this means that they look forward to see how they will behave for different histories and signals that they get. For the Sequential Buying condition, there is less mismatch

between the information sets, the player sees the history they will be at and maximizes the expectation given the distribution of signals. If we look at choosing reporting truthfully against withholding, the player should choose reporting truthfully when

$$r + s + [E[Acc|T] - E[Acc|W]] > \varepsilon_{Wm} - \varepsilon_{Tm} \tag{9}$$

Because the player expects their choices, they will choose to buy when

$$\sum_{\text{possibilities}} \max[U_C^B] - \max[U_C^{DB}] > 0.$$

These shocks are type I extreme value distributed with scale $1/\lambda$, where λ is the population's precision parameter. Then, the limiting case of $\lambda = 0$ has a player choosing uniformly among the options available at each decision: each of T, W, and M a third of the time after buying, and each of F and Q half of the time after not buying.

The logit model allows for the dual role of preferences first in determining buying and then in determining reporting. Concretely, we draw psychological parameters $r \geq 0$ and $s \geq 0$ and a precision parameter λ. Then, we simulate predicted behavior at each possible history.

To estimate the model we use a maximum likelihood estimation. Denote $f(y|\theta)$ the probability of observing the given behavior in this round given population parameters θ. Then, the behavior given their $\theta = \langle r, s, \lambda \rangle$ is the product of likelihoods for each observation.

We use a quantal response equilibrium to handle the strategic interaction of players. This allows us to have players noisily playing their strategies and being jointly induced by the noisy play of others to play their current strategies. To solve for our equilibrium we need to solve the Bayesian equilibrium. To have a Bayesian equilibrium, for some set of parameters, we have to jointly have beliefs of past players inducing beliefs that will lead to the same strategies given histories. To numerically converge we have to jointly estimate beliefs about past behavior and beliefs about future behavior. These need to be consistent. To understand the strategy take player 3 for instance, we calculate the new expectation of behavior by past players (players 1 and 2) and future players (players 4 and 5). This gives us a best response to any history by player 3 (we do this same exercise for the other players). Then, players are moved from their old σ_{old} towards the best response in a damped manner ($\sigma_{new} = 0.4\sigma_{br} + 0.6\sigma_{old}$),[10] iterating until strategies stop moving. This will converge to a quantal response equilibrium.

This game may have multiple equilibria because of its complexity but resampling the space from different starting points, we do not find different equilibria, giving us confidence that other equilibria are not prevalent. We assume that players are playing with the entire population when they are playing quantal response, not the session that they actually played with. When estimating heterogeneous and homogeneous parameters we find little difference between the parameters that we estimate (this choice is motivated by the computational cost of drawing without replacement from sessions: when bootstrapping these costs add up).

[10] 0.4 is the damping parameter, it controls how quickly the new strategies move. If it is too low the system will take longer to converge; too high and the system may overshoot the equilibrium.

B. Identification

We have already learned a decent amount about our identification strategy with the choice behavior of players in our model. We will comment on a few more parts of the strategy, particularly our precision parameters and for our later robustness, how we identify how much subjects care about their instrumental incentives (α).

First, we think about what is identified at the lower level report decisions. At the report decision after buying a signal, telling the truth will always weakly dominate (excluding the error term). Then, withholding implies that $r+s$ is small and that the error term is large (this is true when r and s are greater than 0). Similarly, choosing to misreport implies that s is large and the error is large. We essentially classify misreports and withholding as errors (this is because $r,s>0$).

At the report decision after not buying a signal, $r-s$ is pitted against λ. If a player fabricates, that means that $r-s$ is larger than the error term. So, either, the precision is low or reporting concerns outweigh sincerity concerns.

At the buying level, players will choose to buy and report truthfully rather than fabricate or stay quiet when $s > c - E[\Delta(T,F)]$ and $r > c - E[\Delta(T,Q)]$. When the expected accuracy gain from a signal is small, this is approximately $c < s$ and $c < r$.

C. Benchmark Estimates

Table 10 reports the homogeneous r and s QRE estimates and Table 11 compares them with the QRE.

In the No Communication expressive model we see a high r and low s. This is consistent with classic expressive voting models where people want to vote and have little aversion to uninformed voting.

In the Signal Sharing expressive model we see s slightly above c (0.263 against 0.25) and r well below c (0.148). This leads players who did not buy a signal to stay quiet when the tally of reports is weak and to report truthfully when they bought one. When the tally is already high and they did not buy a signal, they will tend to fabricate.

In the Sequential Buying expressive model we see s and r both well below c (0.138 and 0.153 against 0.25). This indicates that players will tend to report truthfully when there is some benefit to it, but will flip a coin between fabricating and staying quiet when they see little benefit to reporting. This is consistent with Tables 7 and 9 where we see a lot of fabrication when the tally is high and more staying quiet when the tally is low.

We see that the MLE is higher for the estimations with more parameters. This is to be expected, you will always generate at least as good of a fit when you relax assumptions. Using AIC and BIC we can see that the model with our expressive motivations is a much better model than the purely instrumental model.[11]

[11] A difference of 10 is considered strong evidence in favor of the model with a lower AIC or BIC.

TABLE 10—HOMOGENEOUS r, s AND λ BY CONDITION (SHOCK-KNOWN CHOICE, POPULATION LEVEL)

					Signal buy rate		Accuracy	
Condition	r	s	λ	Log-likelihood	Model	Observed	Model	Observed
No Communication	0.381 [0.057]	0.071 [0.023]	7.630 [1.095]	−819.1	37.2%	37.2%	66.2%	72.7%
Signal Sharing	0.148 [0.031]	0.263 [0.049]	6.207 [0.807]	−871.9	41.0%	41.3%	76.6%	70.7%
Sequential Buying	0.153 [0.031]	0.138 [0.035]	5.873 [0.540]	−968.3	38.6%	39.5%	73.8%	80.7%

Notes: Bootstrap standard errors, from resampling subjects within a condition, are in brackets.

TABLE 11—PARAMETER ESTIMATES AND INFORMATION CRITERIA: HOMOGENEOUS r, s, λ MODEL AGAINST THE QRE

								Signal buy rate		Accuracy	
Condition	Model	r	s	λ	Log-lik.	AIC	BIC	Model	Observed	Model	Observed
No Communication	Homogeneous	0.381	0.071	7.630	−819.1	1644.1	1658.3	37.2%	37.2%	66.2%	72.7%
	QRE			5.537	−1146.8	2295.6	2300.3	30.8%	37.2%	59.9%	72.7%
Signal Sharing	Homogeneous	0.148	0.263	6.207	−871.9	1749.9	1763.7	41.0%	41.3%	76.6%	70.7%
	QRE			6.306	−1002.7	2007.3	2011.9	29.0%	41.3%	64.6%	70.7%
Sequential Buying	Homogeneous	0.153	0.138	5.873	−968.3	1942.7	1956.6	38.6%	39.5%	73.8%	80.7%
	QRE			5.580	−1022.3	2046.5	2051.1	31.8%	39.5%	66.1%	80.7%

D. Robustness

For our robustness checks, we will use a nested model. This model specifies a shock at the lower level report decisions that is unknown at the upper level buy decision. There is also a shock at the buy decision that can affect buy behavior. We will still call the precision of the lower level choice λ and call the precision of the upper level choice μ. This model is used because it is far more computationally tractable than the shock-known model.[12] We see very similar results, with lower AIC and BIC for the shock-known model (largely driven by the extra parameter, μ in the nested model).

For the heterogeneous models, we allow different players to have different r and s parameters. We will index players with m and index the round with t. This allows us to estimate a player's heterogeneous reporting preference, r_m, and sincerity utility, s_m, by looking at the generated log likelihood of their observed behavior across all rounds.[13] We then use these estimates to look at the distribution of r_m and s_m across players. Model 5 leaves r_m and s_m unrestricted. In model 6, consistent with our theory that subjects want to report and want to be sincere, we bound r and s to be positive. This assumption means that players that misreport and withhold signals contribute to lowering the estimated choice precision. For both models Table 12 reports the mean of the subject-level estimates and their standard deviation across subjects; this standard deviation includes estimation error.

For model 7 we add α to scale the instrumental payoff. This allows us to see how sensitive players are to the instrumental payoff. Lastly, in model 8 we unbind the utility that a player enjoys for the different reporting options. In our model we

[12] The cost of computing the shock-known model is significantly higher and, when bootstrapping, the computational burden is significant.

[13] Specifically the probability of observing a player's behavior is the product of the probabilities of observing their behavior in each round $\prod_{t=1}^{15} P(C_{mt}|r_m, s_m, \sigma)$.

have imposed a structure on the reporting utilities, for instance that withholding and staying quiet provide the same utility to a player. In model 8 we set the utility that a withholding player enjoys to be 0 and allow a δ_C to be estimated for each of the other reporting options. This allows us to see if the structure that we have imposed is consistent with the data. We do not report a μ for model 8. Table 13 compares these utilities with those implied by r and s in model 4. We have to multiply the precision parameter to compare the models because the size of their parameters is not comparable except when controlling for noise; dividing by λ gives utility units. Model 8 was estimated relative to truthful reporting and is re-expressed relative to withholding, which changes no choice probability. A utility estimated at its bound of ± 10 has no meaningful standard error.

Standard errors in Tables 12 and 13 are bootstrapped by resampling subjects within a condition, with 100 to 200 replicates per model. We compute AIC $= 2k - 2\ell$ and BIC $= k \ln N - 2\ell$, where ℓ is the maximized log likelihood, k is the number of estimated parameters (including each subject's own in the heterogeneous models), and N is the number of player-rounds (825 in No Communication and 750 in each of Signal Sharing and Sequential Buying). The predicted number of signals bought and the predicted accuracy are those implied by the fitted equilibrium, with the group deciding by the majority of the reports.

We see that, in general, the models that allow for expressive utility do very well at matching the rate of signals bought. This is to be expected for a model fitted to choice data. They perform much worse at matching accuracy which they were not fitted on. Most models have similar predicted accuracy to one another and either over or undershoot accuracy.

In No Communication model 7 we see that α is estimated to be 0.277, indicating that players are not very sensitive to the instrumental payoff. In model 8, we see that the log likelihood of the unstructured model is the same as the structured model, indicating that the structure we imposed fits the data well. The model with the lowest BIC is model 2 in No Communication and Signal Sharing and model 8 in Sequential Buying.[14]

In Signal Sharing and Sequential Buying models we see that they estimate $\alpha \approx 1$. This is consistent with the idea that players are more sensitive to the instrumental payoff in these conditions. In Table 13 we see that binding the reporting utilities to r and s provides a worse fit than the unstructured model. In particular, the unstructured model estimates the utility of misreporting as being much lower than the utility of misreporting in the structured model (-0.87 against 1.01 in Signal Sharing and -1.44 against 0.92 in Sequential Buying, in units of λ times utility). Otherwise their utility predictions are fairly close, within 0.4 of each other. The structured model has the lower AIC and BIC in No Communication, and the unstructured model has the lower AIC and BIC in Signal Sharing and Sequential Buying.

[14] Model 5 has the lowest AIC in Signal Sharing and Sequential Buying (model 6 in No Communication), but these models add a lot of parameters, creating fears of overfitting. In general BIC guards more against overfitting.

TABLE 12—MODEL COMPARISON

	Precision		Expressive Tastes				Instr. Weight	Fit				Predicted	
Model	λ	μ	r		s		α	k	Log Lik.	AIC	BIC	# Sigs.	Accuracy
NO COMMUNICATION													
Shock Known													
(1) Instr.	5.54							1	−1146.8	2295.6	2300.3	1.54	59.9
	[0.81]												
(2) Instr. & Expr.	7.63		0.381		0.071			3	−819.1	1644.1	1658.3	1.86	66.2
	[1.10]		[0.057]		[0.023]								
Nested Shocks													
(3) Instr.	6.83	4.23						2	−1140.7	2285.3	2294.8	1.71	62.6
	[0.93]	[1.27]											
(4) Instr. & Expr.	7.92	7.05	0.367		0.066			4	−819.0	1646.0	1664.9	1.86	66.3
	[2.14]	[168.58]	[0.112]		[0.054]								
(5) With Het. Expr.	4.84	357.19	5.917	(3.721)	0.039	(3.751)		112	−502.2	1228.4	1756.5	1.86	65.0
	[0.33]	[106.24]	[0.499]	[0.226]	[0.443]	[0.590]							
(6) With Het. Expr. ≥ 0	4.89	375.57	6.205	(3.778)	0.832	(2.592)		112	−501.9	1227.9	1756.0	1.86	65.2
	[0.08]	[116.11]	[0.500]	[0.221]	[0.347]	[0.658]							
(7) With Instr. Weight	28.62	2.60	0.102		0.018		0.277	5	−819.0	1648.0	1671.6	1.86	66.3
	[6.50]	[0.84]	[0.021]		[0.011]		[0.123]						
(8) With Report-Level Utils[a]	2.16							6	−819.0	1650.0	1678.3	1.86	66.3
	[0.30]												
Observed												1.86	72.7
SIGNAL SHARING													
Shock Known													
(1) Instr.	6.31							1	−1002.7	2007.3	2011.9	1.45	64.6
	[0.60]												
(2) Instr. & Expr.	6.21		0.148		0.263			3	−871.9	1749.9	1763.7	2.05	76.6
	[1.00]		[0.029]		[0.055]								
Nested Shocks													
(3) Instr.	14.36	2.58						2	−959.2	1922.3	1931.6	1.97	75.1
	[1.87]	[0.88]											
(4) Instr. & Expr.	5.72	9.20	0.177		0.301			4	−874.2	1756.3	1774.8	2.07	77.1
	[1.28]	[5.32]	[0.038]		[0.073]								
(5) With Het. Expr.	4.17	25.98	0.162	(0.228)	3.015	(4.563)		102	−607.5	1419.0	1890.2	2.07	76.6
	[0.88]	[9.49]	[0.035]	[0.057]	[0.683]	[0.450]							
(6) With Het. Expr. ≥ 0	4.21	28.05	0.222	(0.117)	3.629	(4.623)		102	−621.9	1447.9	1919.1	2.08	77.1
	[0.68]	[7.01]	[0.020]	[0.016]	[0.636]	[0.247]							
(7) With Instr. Weight	5.73	9.20	0.177		0.301		0.998	5	−874.2	1758.3	1781.4	2.07	77.0
	[44.08]	[6.27]	[0.059]		[0.111]		[0.417]						
(8) With Report-Level Utils[a]	2.24							6	−866.4	1744.7	1772.5	2.07	77.1
	[1.91]												
Observed												2.07	70.7
SEQUENTIAL BUYING													
Shock Known													
(1) Instr.	5.58							1	−1022.3	2046.5	2051.1	1.59	66.1
	[0.50]												
(2) Instr. & Expr.	5.87		0.153		0.138			3	−968.3	1942.7	1956.6	1.93	73.8
	[0.70]		[0.041]		[0.051]								
Nested Shocks													
(3) Instr.	8.65	3.36						2	−1012.7	2029.4	2038.7	1.86	71.7
	[1.42]	[1.09]											
(4) Instr. & Expr.	5.33	7.03	0.172		0.160			4	−968.2	1944.5	1963.0	1.93	73.7
	[0.65]	[1.93]	[0.042]		[0.051]								
(5) With Het. Expr.	4.26	23.46	0.268	(2.873)	0.498	(3.240)		102	−644.2	1492.4	1963.6	2.02	75.2
	[0.47]	[4.02]	[0.409]	[0.549]	[0.455]	[0.646]							
(6) With Het. Expr. ≥ 0	4.61	15.61	0.739	(1.993)	1.024	(2.682)		102	−703.7	1611.5	2082.7	2.05	76.1
	[0.54]	[3.10]	[0.254]	[0.531]	[0.406]	[0.740]							
(7) With Instr. Weight	5.33	7.01	0.171		0.160		1.005	5	−968.2	1946.5	1969.6	1.93	73.7
	[0.56]	[2.52]	[0.041]		[0.040]		[0.183]						
(8) With Report-Level Utils[a]	0.32							6	−942.7	1897.4	1925.1	1.95	74.5
	[0.26]												
Observed												1.97	80.7

Notes: Each row is a model estimated by maximum likelihood on the data of one condition. Instr. models set $r = s = 0$; Instr. & Expr. models add a utility from reporting (r) and from sincerity (s). Indented rows each change one feature of model 4. Bootstrap standard errors are in brackets. For models 5 and 6, the r and s columns report the cross-subject mean with the standard deviation in parentheses. k is the number of estimated parameters. # Sigs. is the mean number of signals bought by a group in a round and Accuracy is the percentage of group decisions that match the true state, both as predicted by the fitted model.

[a] The report-level utilities are reported in Table 13.

TABLE 13—PER-REPORT UTILITY COMPARISON

	No Communication		Signal Sharing		Sequential Buying	
	δ	r,s	δ	r,s	δ	r,s
Utility Relative to Withholding times λ						
δ_W	0†	0†	0†	0†	0†	0†
δ_R	4.008	3.435	2.639	2.736	1.740	1.771
	[1.680]	[0.556]	[0.434]	[0.364]	[0.287]	[0.380]
δ_M	2.259	2.909	−0.870	1.013	−1.435	0.916
	[1.747]	[0.310]	[1.595]	[0.215]	[0.395]	[0.209]
δ_F	4.284	2.909	1.314	1.013	1.208	0.916
	[1.696]	[0.310]	[2.973]	[0.215]	[0.256]	[0.209]
δ_Q	1.941	0.526	2.086	1.723	1.205	0.855
	[1.707]	[0.334]	[2.929]	[0.213]	[0.276]	[0.224]
λ	2.16	7.92	2.24	5.72	0.32	5.33
	[0.30]	[2.14]	[1.91]	[1.28]	[0.26]	[0.65]
Fit						
k	6	4	6	4	6	4
Log Lik.	−819.0	−819.0	−866.4	−874.2	−942.7	−968.2
AIC	1650.0	1646.0	1744.7	1756.3	1897.4	1944.5
BIC	1678.3	1664.9	1772.5	1774.8	1925.1	1963.0
Predicted and Observed						
Sigs. Bought	1.86	1.86	2.07	2.07	1.95	1.93
Observed	1.86	1.86	2.07	2.07	1.97	1.97
Accuracy	66.3	66.3	77.1	77.1	74.5	73.7
Observed	72.7	72.7	70.7	70.7	80.7	80.7

Notes: Utility of each report option, δ_C, relative to withholding, multiplied by the precision λ. The δ columns estimate each utility separately (model 8 of Table 12); the r,s columns impose $\delta_R = r+s$, $\delta_M = \delta_F = r$ and $\delta_Q = s$ (model 4 of Table 12). Bootstrap standard errors are in brackets.
† Fixed at 0.

VI. Conclusions

Consistent with past expressive voting literature, our model estimates a high desire to vote and a small aversion to uninformed voting, capturing the behavior of subjects in our No Communication condition very well. Applying a similarly structured desire to report and desire to report sincerely, our model also captures the behavior of subjects in our conditions with communication well. The one place it falls short is that it assigns too large a utility to misreporting signals. This likely stems from the fact that misreporting is much closer to lying than the other types of reporting. Together, a desire to report to the group, a desire to report sincerely, and a desire not to misreport signals form a small and tractable model of behavior in committees.

This model allows us to think about how committees change as communication and information become cheaper. A decrease in the cost to communicate, from social media or other transformative technologies, amounts to an increase in the utility from reporting. Then, a decrease in the cost of communication makes people report truthfully and fabricate more often. What decides which of the two grows faster is the

incentive to be sincere, not the incentive to report. When the utility from sincerity is small, the proportion of fabrication in the reports grows.

In contrast, a decrease in the cost of information leads to more truthful reporting with no change in fabricating. This is true in both the instrumental and the expressive models, but only the expressive model predicts that members will keep acquiring information even as the group grows.

Because an expressive desire is so powerful at inducing committee members to acquire information, a committee designer can depend less on reputation and more on expressive motivations. However, these motivations do create some unique issues. Just as we should guard against fabrication and herding in committees with strong reputational mechanisms, we need to guard against them in non-reputational committees. Our Sequential Buying condition shows one way to do so. Subjects in every condition buy signals at similar rates, so Sequential Buying groups are more accurate not because the mechanism coordinates signal buying more efficiently, but because it makes the harm of fabricating more apparent to subjects. The best committee may not be the one that most motivates members to acquire information, it may be the one that makes members most reticent to fabricate.

There is no ability for subjects in our experiment to be caught in a fabrication. This is likely an important incentive in real world committees, where two members go toe to toe on a topic and eventually come to a consensus on who is correct. Surely potential embarrassment and glory are important to how people decide whether to report only when informed. Future experiments should study this mechanism.

Appendix

A. Participant Instructions

Page 1: Welcome

Today you will make decisions and interact with other participants. Your choices determine how much you and your group are paid at the end of the experiment.
Please turn off your cellphone. Do not talk to or look at other participants' screens during the experiment.

Experiment Structure

1. **Instructions** with comprehension questions.
2. **15 Rounds** of the Decision Task.
3. A short **Survey**.

Payment

You will receive a **$5 show-up fee** plus earnings from each round of the Decision Task.
We use **points** as currency. Every **1000 points = $1**.

Groups

You are in a group of **5 players**. Your group changes each round. There is a new, randomly formed groups at the start of each round.

Next

Page 2: The Decision Task

Each Round Has Three Stages

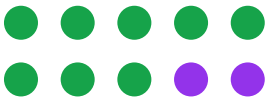
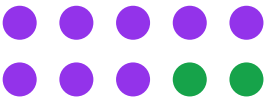
(1) Signal Purchase → (2) Communication → (3) Vote

The Group Color

At the start of every round the computer secretly chooses either **Green** or **Purple** as the true group color. Both colors are **equally likely (50 / 50)**. Nobody sees the true color directly — you have to infer it from signals.

Signals

Before voting you may pay **250 points** to purchase a signal about the true group color. The signal is a colored ball drawn from a virtual bag. Because there are more balls of the correct color in the bag, the signal gives you a better chance of guessing correctly.

	
When true color is **Green** 8 green · 2 purple	When true color is **Purple** 8 purple · 2 green

The signal shows the **correct color 80%** of the time and the wrong color 20% of the time. You can also choose to skip the signal for free, keeping your points.

Voting and Payoff

After the signal (and any communication), all 5 players vote: **Green**, **Purple**, or **Do Not Vote**.
The **majority wins**: the color with more votes is the group decision. In the case of a tie, the decision is made randomly (50% of the time green, 50% of the time purple).
If the group decision matches the true color, **every player earns 1000 points**. If they do not match, no players gain points that round.
Signal costs are deducted regardless of whether the group is correct.

Key numbers to remember
Signal accuracy: **80%** · Signal cost: **250 points** · Correct group: **1000 points each** · Group size: **5 players**

Next

Page 3: Signal Purchase — Demo

[Interface shown on this page:]
Your Information This Round
—

Signal Purchase
Would you like to purchase a signal about the group's true color?
A signal is correct with **80% probability**.
Cost: **250 points**
Buy Signal Don't Buy

[Popups, shown one at a time. The demo draws a Green signal with probability 80% and a Purple signal with probability 20%; [color] below is replaced by the drawn color.]

1. Let's walk through the **Signal Purchase** stage of each round.
   Next →
2. The **Your Information This Round** panel (highlighted) will show you everything you know this round — including any signal you purchase. Right now it is empty.
   Next →
3. The **Signal Purchase** area (highlighted below) asks whether you want to pay **250 points** for a signal. The signal ball will be drawn from the urn — correct color 80% of the time.
   OK — let me try
4. Click **Buy Signal** to see what a signal looks like.
5. *[After Buy Signal is clicked, the panel shows "Signal: [color]" and a [color] ball.]*
   Your signal appeared in the **Your Information** panel — a **[color]** ball. This means that the group color is more likely to be [color]. **Remember: this is correct only 80% of the time.**
   Next →
6. You could also choose **Don't Buy** to keep your points and not get a signal.
   Continue →

Page 4: Signal Calculator — Demo

[Interface shown on this page:]

Signal Calculator

Signal 1	Green	Purple
Signal 2	Green	Purple
Signal 3	Green	Purple
Signal 4	Green	Purple
Signal 5	Green	Purple

Reset all



Green: **50.00%** Purple: **50.00%**

[The bar shows the posterior probability of each color given the entered signals, starting from 50/50 and using 80% signal accuracy.]

[Popups, shown one at a time:]

1. This is the **Signal Calculator**. It helps you combine signals from multiple players to estimate the probability that the true color is Green or Purple.
   Next →
2. Each row represents one signal. Click **Green** or **Purple** to enter a signal. Click the active color again to remove it. Try entering a **Green** signal in row 1 now.
3. *[After a Green signal is entered in row 1:]*
   After one **Green** signal the probability shifts from 50/50 to **80% Green** — exactly the signal accuracy. Try adding more signals and watch the bar move.
   Next →
4. The **bar** (highlighted) always sums to 100%. Green fills from the left; purple fills from the right. The more signals agree, the more confident the bar becomes.
   Next →
5. **Reset all** clears every signal so you can start a fresh calculation. Click it now, then press Continue.
   Continue → *[enabled after Reset all is clicked]*

Page 5: Overview

Round Structure

Players take turns. Before the next player gets their turn, each player **decides about their signal and sends a message**.

1. **Player 1 goes first.**
   Player 1 sees the Buy / Skip screen, decides whether to buy a signal. Then they see their signal result. Then they send one message to the group. While Player 1 is active, Players 2–5 see the chat log filling in.
   [P1 — deciding] [P2] [P3] [P4] [P5]
2. **Then Player 2, then 3, then 4, then 5 do this process.**
   By the time it is your turn you will have already read the messages from all earlier players.
   [P1 √] [P2 √] [P3 — deciding] [P4] [P5]
   [Example chat:]
   Player 1: Hi everyone.
   Player 2: Good luck this round.
3. **After Player 5 sends their message, everyone votes simultaneously.**

Key point: You send exactly **one message**. You do this after seeing the messages from all earlier players, and before the later players' messages and signal decisions. Please refrain from using profanity or identifying yourself.

[Next]

Page 6: Voting — Demo

[Interface shown on this page:]
Your Information This Round
Signal: Green *[green ball]*

Chat log:
Player 1: Hi everyone.
Player 2: Good luck this round.
Player 3: Hello all.
Player 4: See you at the vote.

Cast Your Vote
Vote for the color you believe is the group's true color, or choose not to vote.

Vote Green | Vote Purple | Do Not Vote

[After a vote button is clicked, the status line reads:] Vote recorded.

[Popups, shown one at a time:]

1. Now let's look at the **voting stage**. This is what you see just before casting your vote.
   Next →
2. The **Your Information** panel (highlighted) shows your signal and all messages sent earlier in the round. By the time voting opens, every player has already shared their signal and reasoning in the chat.
   Next →
3. Click **Vote Green**, **Vote Purple**, or **Do Not Vote**. The color with more votes becomes the group's decision. **Try clicking a vote button now.**
4. Your vote is sent. Once all 5 players have voted, the results appear — you will see the vote tally and whether the group was correct.
   Finish Demo →

Page 7: Comprehension Check

Answer the questions below. If any answer is wrong you will be shown a hint and asked to try again.

Signal Accuracy

What is the probability that a purchased signal shows the correct group color?

- ◯ 50%
- ◯ 80%
- ◯ 30%

[Hint shown if answered incorrectly:] A signal shows the correct color 80% of the time.

Group Decision Rule

How is the group decision made?

- ◯ The color that gets the most votes wins.
- ◯ A randomly chosen player decides for the group.
- ◯ All players must agree on a color.

[Hint shown if answered incorrectly:] The group picks the color that receives the most votes (simple majority).

Your Treatment

When does Player 3 get to decide about their signal?

- ◯ At the same time as every other player.
- ◯ After all other players have already voted.
- ◯ After all players with a lower number have decided about their signal and sent a message.

[Hint shown if answered incorrectly:] Each player goes in order — Player 3 waits for Players 1 and 2 to finish.

Next

[Correct answers: 80%; The color that gets the most votes wins.; After all players with a lower number have decided about their signal and sent a message.]

B. Instrumental Theory

We briefly present the theoretical results for the No Communication and Signal Sharing conditions (we will call them the Simultaneous Buying conditions to group them). The two Simultaneous Buying conditions have identical equilibria which are very similar to those presented in Großer and Seebauer (2016), Bhattacharya, Duffy, and Kim (2017), and Elbittar et al. (2020). Then, we present our subgame perfect Nash equilibrium of the Sequential Buying condition.

1. Simultaneous Signal Buying Conditions (No Communication and Signal Sharing)

Under *No Communication Voting Strategies* we show that the No Communication condition induces players that bought signals to vote for that color and players that did not buy signals to abstain. Then, under *Signal Sharing Reports and Votes*, we show that the Signal Sharing condition induces players to truthfully report signals, then vote to choose the alternative that is supported by those signals (these are both implied by the common value environment with majority voting). We show that for any signal buying behavior and signal realizations, the groups in the two conditions will always choose the same alternative. This implies that the signal buying behavior will be identical in the two conditions. Using this insight, under *Signal Buying*, we compute the identical symmetric signal buying Nash equilibrium for these two conditions.

No Communication Voting Strategies Figure 1 shows the timing of the No Communication condition: committee members all buy signals simultaneously, then vote. Because subjects are rematched every round and they cannot communicate before getting signals, we only present symmetric equilibria in which every subject buys a signal with the same probability φ. This is motivated by the repeated rematching and the subjects' relative inability to coordinate on sophisticated equilibria.[15]

Uninformed Voters: The voting stage for a member that did not buy a signal is the problem studied by Feddersen and Pesendorfer (1996). They show that absent partisan voters, an uninformed voter strictly prefers to abstain even though voting is free. A vote only affects the outcome when it makes or breaks a tie. Since the uninformed voter's ballot carries no information, they will either turn an informed majority decision into a tie or turn a tie into an uninformed decision. This is strictly worse than not voting.

Informed Voters: Austen-Smith and Banks (1996) show that if a majority rule applied to signals is the most efficient way to aggregate information, then sincere voting is a Nash equilibrium. Because our signals are equally precise and independent and the prior is symmetric, the posterior favors the color that had the majority of signals. So, majority rule is the most efficient aggregation rule. We will show the

[15] Koriyama and Szentes (2009) describe an equilibrium where players coordinate on some people always acquiring signals, some people buying signals with probability φ, and some never buying. This equilibrium would require coordination that our subjects should not be able to achieve.

intuition of this by using pivotality. Conditional on being pivotal, the others' ballots are tied, so the voter's own signal decides the outcome. This means that you should vote for the color indicated by your signal because you might miss out on the chance to choose the correct color.

Signal Sharing Reports and Votes Figure 2 shows the timing of this condition: players buy signals simultaneously, then communicate their signals to the committee. After communication is done, members vote simultaneously. For the same reasons that we focus on a symmetric Nash equilibrium in the No Communication condition—the relative inability to coordinate before buying signals and the random rematching in our experiment—in the Signal Sharing condition we also only present symmetric equilibria.

We follow similar logic as we do in the No Communication condition. Instead of considering whether voters will vote truthfully, we need to consider whether players will truthfully report. If we assume that everyone votes truthfully, then reporting untruthfully will lead to a worse decision in the event that signal changes which color is supported by the preponderance of the signals. A player is indifferent between lying and truth-telling when the report cannot change the outcome, and weakly prefers truth-telling otherwise (lying is weakly dominated).

Votes follow the majority of the reported signals. So, pivotal voters will always vote in the direction of the signals (voting for the supported alternative is weakly dominant). We focus on signal reporting and voting strategies that survive the elimination of weakly dominated strategies; these strategies translate bought signals into a choice for the alternative supported by the aggregate signal.

Signal Buying We have seen that both conditions truthfully transform signals into a group decision. For any set of signals and their realizations, the two conditions will come to the same aggregate decision.

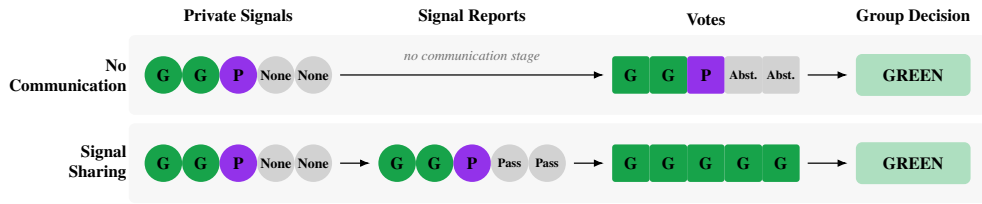


FIGURE 8. NO COMMUNICATION AND SIGNAL SHARING TRANSLATE SIGNALS INTO THE SAME DECISIONS

Figure 8 shows how signals map into the group decision in the two Simultaneous Buying conditions. Because signals in the two conditions translate into the same aggregate decision in our theory, we only need to compute the symmetric Nash equilibrium for the No Communication condition and it will be applicable to both games.

A truthful vote by a No Communication voter will be pivotal when:

- Their vote changes the vote split from $j:j$ to $j+1:j$ in favor of their signal. This is an improvement on the group's previous choice because it transforms a coin flip into a group choice for the signal's color.
- Their vote changes the vote split from $j+1:j$ to $j+1:j+1$ (the group makes a random choice). This is not beneficial because when they are pivotal this way, both alternatives are equally likely to be true.

No Communication voters will buy signals with the goal of turning a tie into a decision in the direction of their signal. This takes the decision from a coin flip to choosing the alternative that has a probability x of being correct. Let m be the number of the other $n-1$ members who buy a signal, each with probability φ. The expected marginal accuracy, E[MA], for a group member of buying a signal is:

$$\mathrm{E[MA]} = \underbrace{(x-0.5)}_{\text{Marg. Acc. | Break Tie}} \cdot \underbrace{\sum_{m=0}^{n-1} \binom{n-1}{m} \varphi^m (1-\varphi)^{n-1-m} \cdot \binom{m}{m/2} [x(1-x)]^{m/2} \mathbb{1}(m \text{ even})}_{\text{Probability Break a Tie}} \tag{10}$$

For $n=5$ (4 other players) a player's E[MA] given the probability that other players buy a signal, φ, is:

$$\mathrm{E[MA]} = (x-0.5)\left[\underbrace{(1-\varphi)^4}_{\text{Nobody Buys}} + \underbrace{12\varphi^2(1-\varphi)^2 x(1-x)}_{\text{2 Players Buy}} + \underbrace{6\varphi^4 x^2(1-x)^2}_{\text{4 Players Buy}}\right] \tag{11}$$

$$\partial \mathrm{E[MA]}/\partial\varphi = (x-0.5)\Big[\underbrace{-4(1-\varphi)^3}_{<0 \text{ at } \varphi=0} + 24\varphi(1-\varphi)(1-2\varphi)x(1-x) + \underbrace{24\varphi^3 x^2(1-x)^2}_{>0 \text{ at } \varphi=1}\Big] \tag{12}$$

Notice that $\partial \mathrm{E[MA]}/\partial\varphi$ at $\varphi=0$ sets everything but the negative leftmost term to zero and at $\varphi=1$ sets everything but the positive rightmost term to zero. Because the derivative is negative at $\varphi=0$ and positive at $\varphi=1$, using the intermediate value theorem, we know that the derivative of E[MA] must be zero at some point. This implies that there is a minimum E[MA] between 0 and 1. E[MA] is strictly convex,[16] so the minimum is unique. For the signal strength that we use in our experiment, $x=0.8$, the minimum $\mathrm{E[MA]}^* \approx 0.034$ at $\varphi^* \approx 0.841$.

When $c < \mathrm{E[MA]}^*$, every player strictly prefers to buy, so $\varphi=1$ is the unique symmetric equilibrium. When $\mathrm{E[MA]}^* < c < \mathrm{E[MA]}(1)$, where E[MA](1) is E[MA] at $\varphi=1$, there are three symmetric equilibria: $\varphi=1$ and the two values of φ

[16] Drop $(x-0.5)$ and let $\theta = x(1-x)$. Then we are left with $12(1-\varphi)^2 + 72\theta^2\varphi^2 + 24\theta(1-2\varphi)^2 \mathbf{-48\theta\varphi(1-\varphi)}$ which is not obviously positive because of the negative bold term. Absorb the last term by completing the square $12(1-\varphi-\mathbf{2\theta\varphi})^2 + \mathbf{24\theta^2\varphi^2} + 24\theta(1-2\varphi)^2$. Each term is nonnegative, and the sum is strictly positive because the first two terms will not vanish together.

that solve $c = \text{E[MA]}(\varphi)$. When $\text{E[MA]}(1) < c < x - 0.5$, the unique symmetric equilibrium is the φ that solves $c = \text{E[MA]}(\varphi)$.[17] At our experimental cost, $c = 0.25$, it is $\varphi \approx 0.046$. When $c > x - 0.5$, nobody buys. Whenever $\varphi = 1$ is an equilibrium ($c < \text{E[MA]}(1) \approx 0.046$), we select it in the comparisons below.

This same behavior will hold in the Signal Sharing condition. Signals are also bought with probability φ. Instead of the signal being translated into a vote, signals are translated into signal reports which are later aggregated into votes.

2. *Sequential Buying*

Figure 3 shows that players one by one buy and see their signals, then have the opportunity to report their signal. If players communicate truthfully, they get to see other players' signal buying decisions and signal realizations before making their own signal buy decision.

We first verify that players will report truthfully and vote to choose the alternative supported by the signals. If players report truthfully, voting against the alternative supported by the signals is detrimental when a voter is pivotal (a weakly dominated strategy). If they know that the preponderance of the evidence is going to be voted on, for the same reason, reporting truthfully is weakly dominant. Then, it is a Nash equilibrium to report truthfully and vote for the alternative supported by the evidence.

Because signal buying depends on previous signal reports, we need to transform the signals collected into the absolute difference in the signals for both alternatives, $k = |\sum_{i \in I} \mathbb{1}(\varsigma_i = \rho) - \sum_{i \in I} \mathbb{1}(\varsigma_i = \gamma)|$. Then, the probability that the group is correct is:

$$\pi(k) = \frac{1}{1 + \left(\frac{1-x}{x}\right)^k} \tag{13}$$

Given truthful reports and majority voting, the group's accuracy is $\pi(k)$. We will analyze how much one more signal changes it. k changes when a committee member buys a signal, either increasing with a confirming signal or decreasing with a disconfirming one. The evolution of k can be thought of as a Markov chain.

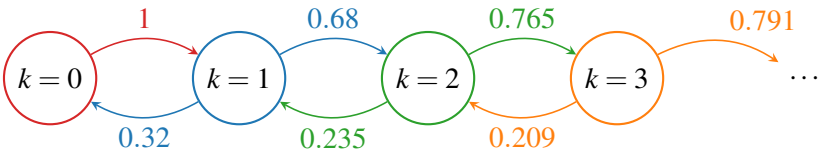


FIGURE 9. MARKOV CHAIN OF k FOR $x = 0.8$

Notes: Arrow labels are the probabilities that the next signal moves the tally k up or down.

Figure 9 shows the transition probabilities for buying a signal when $x = 0.8$. Call the probability that the group gets a confirming signal $q(k)$.[18] When $k \geq 1$, $q()$ is

[17] E[MA] falls from $x - 0.5$ to E[MA]^* on $[0, \varphi^*]$ and stays below $\text{E[MA]}(1)$ on $[\varphi^*, 1]$, so there is exactly one root.

[18] $q(k) = \pi(k)x + (1 - \pi(k))(1 - x)$: when the tally is correct, another correct signal will bolster it and when the tally is incorrect, an incorrect signal will bolster the tally.

an increasing function of k.[19] The same is not true for the expected accuracy after another signal, $\nu(k)$. For $k \geq 1$, any expected information gain from additional signals is 0 ($\nu(k) \equiv q(k) \cdot \pi(k+1) + (1-q(k)) \cdot \pi(k-1) = \pi(k)$). This is a property of Bayesian estimates of discrete, symmetric signals with a symmetric prior; because they incorporate all available information, after the first signal they remove any *expected* information gains from additional signals. An additional signal bought at $k \geq 1$ will never increase the group's expected accuracy on its own, as the following proposition states.

PROPOSITION 1: *Suppose that no player buys a signal after the current one, then*

(i) A signal bought at $k = 0$ raises the group's expected accuracy by $x - 0.5$.

(ii) A signal bought at $k \geq 1$ leaves the group's expected accuracy unchanged: $\nu(k) = \pi(k)$.

The intuition of this proposition is gained by looking at the last and second to last players. Because the last player has the final say, they will only buy a signal that will increase the accuracy of the judgment. Proposition 1 tells us that this can only be the case when the group is at $k = 0$, the uninformative prior. Now, think about the second to last player: they will not buy when $k = 0$ because they can rely on the last player to buy. But they might be willing to buy a signal when $k = 1$, because if they get a disconfirming signal, they know that the last player will buy a signal. This erases the potential downside of buying a signal, which makes buying worthwhile. The formal proof of Proposition 1 is in Appendix C.

We now describe the subgame perfect Nash equilibrium of this game. It combines the signal Markov chain with player i's buy strategy $\sigma_i(k) \in \{0,1\}$, which equals 1 if player i buys a signal at tally k. Let $W(i,k)$ be the probability that the group will be correct at the end of the game when player i is about to move at tally k, the common continuation value. The terminal common continuation value is $W(n+1,k) = \pi(k)$, just the accuracy of the group after the last player. Let $B_i(k) = q(k)W(i+1,k+1) + (1-q(k))W(i+1,k-1)$ be the common continuation value if player i buys a signal. The common continuation value then satisfies the recursion $W(i,k) = \sigma_i(k)B_i(k) + (1-\sigma_i(k))W(i+1,k)$, and player i gets payoff:

$$U_i(k) = \begin{cases} B_i(k) - c & \text{if Buy,} \\ W(i+1,k) & \text{if Don't Buy.} \end{cases} \tag{14}$$

A profile $\boldsymbol{\sigma}$ is a subgame perfect Nash equilibrium if players buy signals just when it is personally beneficial: for every i and k, $\sigma_i(k) = \mathbb{1}[B_i(k) - W(i+1,k) \geq c]$.

THEOREM 1: *Let $n-i$ denote the number of players left to move after the current player. Suppose that every player $j > i$ buys when $k \leq n-j$ and $k \equiv n-j \pmod 2$. Then player i's expected increase in the common continuation value from buying, $B_i(k) - W(i+1,k)$,*

$$\textit{is positive} \quad \Longleftrightarrow \quad k \leq n-i \ \textit{ and } \ k \equiv n-i \pmod 2,$$

If at least one condition does not hold, $B_i(k) - W(i+1,k) = 0$.

[19] Because $\pi(k)$ is increasing in k, as k increases $q(k)$ increases too, approaching x.

Theorem 1 says that player i might buy a signal if $k \leq n-i$, k, $n-i$ are both odd or both even, *and* c is low enough. Otherwise, they will not buy a signal. We check our logic with the last player ($n-i=0$), they have gains from buying only when $k=0$ (this is exactly the condition outlined in Theorem 1). The second to last ($n-i=1$) player will only buy when $k=1$ (again Theorem 1 holds). The argument is extended by looking at the third to last player ($n-i=2$):

$k=0$ They might buy a signal because they expect the second to last player to buy a signal at $k=1$. This will sometimes lead to terminal $k=2$ and sometimes to a terminal $k=1$.

$k=1$ They will not buy a signal because they can always depend upon the next player to buy a signal. This will lead to the same outcome as buying, so they will free-ride on the next player.

$k=2$ They might buy a signal because they expect the next player to buy a signal at $k=1$. If they get a confirming signal, they will get a terminal $k=3$. If they get a disconfirming signal, the next player will buy a signal, sometimes leading to a terminal $k=2$.

$k \geq 3$ When $k \geq 3$ the player knows that, in the event of a disconfirming signal, later players will not buy signals so Proposition 1 binds.

A parity argument explains this pattern. Every signal moves the tally by exactly one (from $k=0$ it moves to 1), so after $n-i$ more signals the tally can be back at 0 only if $k \leq n-i$ and $k \equiv n-i \pmod 2$, which is exactly the buy cell. The proof shows that this is no coincidence:

$$B_i(k) - W(i+1,k) = (x-0.5) \cdot \Pr(\text{the tally is 0 after } n-i \text{ signals, starting from } k).$$

Beyond the first signal, extra information is a martingale (Proposition 1), so a signal is only valuable through the chance that the group ends up tied when the last player moves, who then gains $x-0.5$. Each earlier player's gain is the next player's expected gain at the tally her signal produces; when the next player would buy at the same tally, the earlier player free-rides. For example, the second to last player at $k=1$ gains $(x-0.5) \cdot 2x(1-x) = 0.3 \times 0.32 = 0.096$, the second threshold in Table 14 below. The formal proof of Theorem 1 is in Appendix C.

TABLE 14—$B_i(k) - W(i+1,k)$ FOR $n=5$ AND $x=0.8$

	Player 1	Player 2	Player 3	Player 4	Player 5
$k=0$	0.031	Free-Ride	0.096	Free-Ride	0.3
$k=1$		0.031	Free-Ride	0.096	Martingale
$k=2$			0.023	No Backstop	Martingale
$k=3$				No Backstop	Martingale
$k=4$					Martingale

Notes: Cell numbers are the expected marginal accuracy to the group from player i buying a signal at a particular k. Cells that are blank are unreachable for players in a 5 player group. Free-Ride means that the next player will always pay the cost so that player is unwilling to buy a signal. No Backstop means that if the player buys a signal they will not induce anyone to buy another signal (by Proposition 1, these players do not buy). Martingale means that buying has zero marginal accuracy and the player is the last. The cells for players 1 and 2 (0.031) assume that player 3 does not buy at $k=2$, which holds for $c > 0.023$; if every later player buys in all of their buy cells ($c < 0.001$), these cells are 0.046.

Table 14 gives the marginal benefits of buying for each of the 5 players at the values of k they can encounter in a game. If the cost is lower than the marginal benefit of buying, a player will buy. For instance, when $0.096 < c < 0.3$, only player 5 will buy a signal (this gives an accuracy of 80%). When $0.031 < c < 0.096$, player 3 will buy a signal and player 4 will buy a signal; player 5 will only buy a signal when the signals of players 3 and 4 contradict each other, which gives an expected accuracy of $0.68(0.9412) + 0.32(0.8) = 0.896$.[20] As costs fall and reach each of the cost thresholds, more players will buy signals.

A peculiarity of this equilibrium is that for $0.096 < c < 0.3$, any group size will buy the same number of signals (in the $0.096 < c < 0.3$ equilibrium, a 1 person and a 100 person group will both have the last player buy a signal). When the cost is $0.031 < c < 0.096$, if the group has at least 3 players, it will always replicate the same equilibrium path: two players buy a signal and a third buys an extra signal if their signals disagree. In general, if a c leads to g group members being willing to buy a signal, then in groups with at least g members the last g members of the group will buy a signal. Because a player's equilibrium strategy depends only on k and the number of players left to move, adding a member to the front of the line leaves the strategies of the original members unchanged. This is similar to the Condorcet Jury Theorem: adding members to a committee never hurts the accuracy of the group's decision, and it improves the decision when the added member buys a signal.

3. *Efficiency*

To examine which of the conditions is most socially efficient we construct a first best measure of efficiency. We assume that a benevolent dictator, who observes the signals as they are bought and decides after each one whether to buy another, chooses the buying rule to maximize total group's welfare: the sum of all players' payoffs (n times the group's accuracy minus the cost of the signals bought).

[20] This is strikingly similar to mammogram evaluation procedures where two radiologists independently evaluate the images for any signs of cancer and when their reports match up, (possible cancer, possible cancer) or (no concerns, no concerns), information gathering stops and the decision is made, but when their reports do not align a third person is brought in to break the tie (Ciatto et al. 2005).

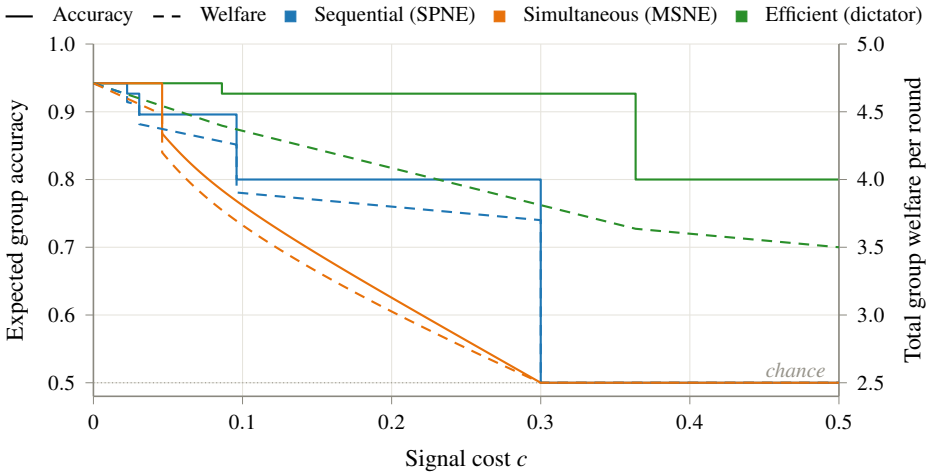


FIGURE 10. SIMULTANEOUS, SEQUENTIAL, AND EFFICIENT LEVELS OF GROUP ACCURACY AND TOTAL GROUP WELFARE ($n = 5$, $x = 0.8$)

For $n = 5$ and $x = 0.8$, Figure 10 presents the expected accuracy and total group welfare for the Sequential Buying equilibrium, the Simultaneous Buying equilibrium, and the benevolent dictator. When $c > 0.3$, it is never individually beneficial to buy a signal, so neither condition buys and both have very low efficiency. The Sequential Buying condition does remarkably well once $c < 0.3$: Sequential Buying's total group welfare is within 11% of the efficient welfare for every $c < 0.3$, and 5% below it at our experimental cost (3.75 versus 3.95 at $c = 0.25$).

For $0.023 < c < 0.046$, the Simultaneous Buying conditions actually do better than the Sequential Buying condition in both accuracy and welfare (given that we select the most efficient $\varphi = 1$ equilibrium in this range). This is a result of the next cost threshold for the Sequential Buying condition not yet being reached and the Simultaneous Buying conditions reaching the range where everyone buying a signal is an equilibrium.

C. *Proofs*

Setup.

- $k = |\sum \rho - \sum \gamma|$ (our running tally of signal reports)
- $\pi(k) = \dfrac{1}{1 + \dfrac{(1-x)^k}{x^k}}$ (the accuracy of the signal tally; $\pi(0) = 1/2$)
- $q(k) = x\pi(k) + (1-x)(1-\pi(k))$ (The probability that a new signal moves k up by 1; $q(0) = 1/2$). A signal at $k = 0$ moves the tally to 1 whichever color it is, so throughout $k-1$ is read as $|k-1|$.
- $W(i,k)$, $B_i(k)$ and $\sigma_i(k)$ are as in the main text, with $W(n+1,k) = \pi(k)$ and $B_i(k) = q(k)W(i+1,k+1) + (1-q(k))W(i+1,k-1)$.
- $G_i(k) = B_i(k) - W(i+1,k)$ (the gain in the group's accuracy from player i buying at tally k).

Proof of Proposition 1. If no one buys after this signal, then a signal bought at $k=0$ always moves the tally up by one, so it raises accuracy from $\pi(0)=1/2$ to $\pi(1)=x$. A gain of $x-0.5$.
Let $D_k = x^k + (1-x)^k$ be the denominator of $\pi(k)$. Then if $k \geq 1$,

$$q(k) = x\pi(k) + (1-x)(1-\pi(k)) = \frac{x^{k+1}}{D_k} + \frac{(1-x)^{k+1}}{D_k} = \frac{D_{k+1}}{D_k}$$

$$q(k)\pi(k+1) = \frac{D_{k+1}}{D_k} \cdot \frac{x^{k+1}}{D_{k+1}} = \frac{x^{k+1}}{D_k}$$

$$1-q(k) = 1 - \frac{D_{k+1}}{D_k} = \frac{D_k - D_{k+1}}{D_k} = \frac{x(1-x)D_{k-1}}{D_k}$$

$$(1-q(k))\pi(k-1) = \frac{x(1-x)D_{k-1}}{D_k} \cdot \frac{x^{k-1}}{D_{k-1}} = \frac{x^k(1-x)}{D_k}$$

Adding them up we get

$$v(k) = \frac{x^{k+1}}{D_k} + \frac{x^k(1-x)}{D_k} = \frac{x^k}{D_k} = \pi(k).$$

□

Proof of Theorem 1. Write $m = n-i$ for the number of players left to move after player i, and say that (i,k) is a *buy cell* if $k \leq m$ and $k \equiv m \pmod 2$. The assertion is that every later player $j > i$ buys exactly in her own buy cells: $\sigma_j(k) = 1$ if and only if (j,k) is a buy cell.

Intuition. The proof rests on three observations.

1. *The last player buys only to break a tie.* By Proposition 1, a signal is worth $x-0.5$ at $k=0$ (it turns a coin flip into an x-accurate decision) and nothing at $k \geq 1$ (it is a martingale). So $G_n(k) = (x-0.5)\mathbb{1}\{k=0\}$.
2. *Value is passed down the line.* If the next player is going to buy at tally k, then my signal at k adds nothing they would not have added themselves (I can free-ride). If they are not going to buy, buying is worthless to them at k, and it is worthless to me unless my signal moves the tally to a place where the next player's purchase is worth something. So, my gain is the *expected gain of the next player at the new tally*.
3. *Induction.* Every signal moves the tally by exactly one (from 0 it moves to 1). Handing the value down m times, my gain is $x-0.5$ times the probability that the tally is back at 0 when the last player moves. That is only possible if $k \leq m$ and the number of steps m has the same parity as k (the buy cell).

Step 1: Since $W(j,k) = \sigma_j(k)B_j(k) + (1-\sigma_j(k))W(j+1,k)$, for every j and k

$$W(j,k) = W(j+1,k) + \sigma_j(k)\,G_j(k). \tag{15}$$

Fix $i \le n-1$. Apply (15) with $j = i+1$ to the two tallies that player i's signal can lead to:

$$\begin{aligned} B_i(k) &= q(k)W(i+1,k+1) + (1-q(k))W(i+1,k-1) \\ &= \underbrace{q(k)W(i+2,k+1) + (1-q(k))W(i+2,k-1)}_{=B_{i+1}(k)=W(i+2,k)+G_{i+1}(k)} + q(k)\sigma_{i+1}(k+1)G_{i+1}(k+1) \\ &\quad + (1-q(k))\sigma_{i+1}(k-1)G_{i+1}(k-1). \end{aligned}$$

(Here $B_{i+1}(k)$ is the value player $i+1$ would get from buying at k, whether or not they actually do.) Applying (15) once more, $W(i+1,k) = W(i+2,k) + \sigma_{i+1}(k)G_{i+1}(k)$. Subtracting,

$$\begin{aligned} G_i(k) = &\underbrace{(1-\sigma_{i+1}(k))\,G_{i+1}(k)}_{\text{free-riding term}} \\ &+ q(k)\,\sigma_{i+1}(k+1)\,G_{i+1}(k+1) + (1-q(k))\,\sigma_{i+1}(k-1)\,G_{i+1}(k-1). \end{aligned} \tag{16}$$

Step 2: Let $\kappa_0 = k$ and let $\kappa_{t+1} = \kappa_t + 1$ with probability $q(\kappa_t)$ and $\kappa_{t+1} = |\kappa_t - 1|$ otherwise. This is the tally if each of the next t players simply drew a signal. Write $\Pr_k$ for probabilities when $\kappa_0 = k$. We show by backward induction on i that, for every $k \ge 0$,

$$G_i(k) = (x-0.5)\,\Pr_k(\kappa_{n-i} = 0)\,. \tag{17}$$

Base case ($i = n$). Here $\kappa_0 = k$, so the right-hand side is $(x-0.5)\mathbb{1}\{k=0\}$, which is $G_n(k)$ by Proposition 1.

Inductive step. Suppose (17) holds for $i+1$. Each step of κ changes it by exactly one, so $\kappa_t \equiv k+t \pmod 2$ and $\kappa_t = 0$ requires $t \ge k$. Conversely, if $t \ge k$ and $t \equiv k \pmod 2$, the path that falls from k to 0 and then bounces between 0 and 1 has positive probability, because $1-q(j) > 0$ for $j \ge 1$ and the move $0 \to 1$ has probability one. Hence $\Pr_k(\kappa_t = 0) > 0$ if and only if $k \le t$ and $k \equiv t \pmod 2$. With $t = n-i-1$, this says $G_{i+1}(k) > 0$ if and only if $(i+1,k)$ is a buy cell, and $G_{i+1}(k) = 0$ otherwise. By hypothesis, $\sigma_{i+1}(k) = 1$ if and only if $(i+1,k)$ is a buy cell, so

$$\sigma_{i+1}(k)\,G_{i+1}(k) = G_{i+1}(k) \quad \text{and} \quad (1-\sigma_{i+1}(k))\,G_{i+1}(k) = 0 \qquad \text{for all } k.$$

The free-riding term in (16) vanishes and the σ_{i+1} drop out of the remaining terms:

$$G_i(k) = q(k)\,G_{i+1}(k+1) + (1-q(k))\,G_{i+1}(k-1).$$

Substituting (17) for $i+1$ and using the first-step decomposition $\Pr_k(\kappa_t = 0) = q(k)\Pr_{k+1}(\kappa_{t-1} = 0) + (1-q(k))\Pr_{|k-1|}(\kappa_{t-1} = 0)$ gives (17) for i. (At $k = 0$ both signals lead to tally 1, and the same identity holds.)

Step 3: By (17) and the parity argument above with $t = n-i = m$, $G_i(k) = B_i(k) - W(i+1,k) > 0$ if and only if $k \le n-i$ and $k \equiv n-i \pmod 2$. Otherwise $\Pr_k(\kappa_m = 0) = 0$, so $B_i(k) - W(i+1,k) = 0$. In particular the gain from buying is never negative, so a player never strictly prefers to buy outside their buy cell and, whenever c is below the gain, strictly prefers to buy inside it. This is the hypothesis for player $i-1$, so the argument extends the equilibrium one player at a time. □

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